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LiH Potential Curves and Wavefunctions for $X\ ^1\Sigma^+$, $A\ ^1\Sigma^+$, $B\ ^1\Pi$, $^3\Sigma^+$, and $^3\Pi$*

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Ab initio multiconfiguration self-consistent-field calculations are reported for the potential curves and electronic wave functions of the states $X\ ^1\Sigma^+$, $A\ ^1\Sigma^+$, $B\ ^1\Pi$, $^3\Sigma^+$, and $^3\Pi$ of LiH. In this calculation, the outer two electrons are correlated, while the 1σ shell, essentially a K shell on Li, is left uncorrelated. The obtained dissociation energies, with the known experimental values in parentheses, are $D_e(X\ ^1\Sigma^+) = 2.411(2.5154)$ eV, $D_e(A\ ^1\Sigma^+) = 1.048$ (1.0765) eV, $D_e(B\ ^1\Pi) = 0.017$ (0.035) eV and $D_e(^3\Pi) = 0.226$ eV.

INTRODUCTION

As the smallest neutral heteropolar molecule, LiH has been a favorite molecule for theoretical investigation. Since a reasonable amount of accurate spectroscopic data is available for LiH, detailed calculations can give a good assessment of a given computational procedure, and at the same time, information which can aid in the understanding of the observed data. Until recently, most theoretical work done on LiH (and on molecules in general) concentrated on describing the ground state and its properties only. The experimental information available goes beyond this and a more complete theoretical study should include the calculation of wavefunctions, potential curves, and properties of ground as well as excited states. It is with this aim that we set out to compute the five valence excited states $X\ ^1\Sigma^+$, $A\ ^1\Sigma^+$, $B\ ^1\Pi$, $^3\Sigma^+$, and $^3\Pi$ of LiH. This allows one to evaluate the capability of the multiconfiguration self-consistent-field (MCSCF) method to yield accurate potential curves, term values and molecular properties for ground and excited states. In addition one can expect to obtain a more detailed understanding of some of the observed anomalies in the LiH spectrum.

A careful spectroscopic analysis of the $A\ ^1\Sigma^+$–$X\ ^1\Sigma^+$ band system of LiH and LiD by Crawford and Jorgensen[1,2] revealed that the $G(v)$ and B_v values of the $A\ ^1\Sigma^+$ state do not show the normally expected decrease with increasing v. Instead they rise initially, before the normal decrease sets in, yielding anomalous negative values for $\omega_e x_e$ and α_e. The same anomaly is observed for the $A\ ^1\Sigma^+$ states of other alkali hydrides. On the basis of their observation, Crawford and Jörgensen concluded that this anomaly must be ascribed to peculiarities of the potential energy curve of the $A\ ^1\Sigma^+$ state alone.[3] Mulliken[4] explained the exceptional shape of the $A\ ^1\Sigma^+$ potential curve as being due to an avoided crossing of the zero-order curves of LiH and Li^+H^-. However, more recently Jenč[5] argued that the anomalous character of the $A\ ^1\Sigma^+$ states of the alkali hydrides should be ascribed to nonadiabatic effects, i.e., a breakdown of the Born–Oppenheimer approximation.

In addition to the $A\ ^1\Sigma^+$–$X\ ^1\Sigma^+$ band system, Velasco[6] identified in 1957 the $B\ ^1\Pi$–$X\ ^1\Sigma^+$ system, characterizing the $B\ ^1\Pi$ state as weakly bound ($D_e = 0.03$ eV). No theoretical study to date has yielded a bound potential curve for this state. Also the corresponding $^3\Pi$ state, which is expected to lie below the singlet state has not as yet been identified.

In the following paragraphs we will give a brief outline of the MCSCF method used in the present calculations, together with some computational details such as basis function choice and selection of configurations. This will be followed by a presentation and discussion of the calculated wavefunctions, potential curves and expectation values of various one electron operators. A more detailed spectroscopic analysis, including the calculation of rotation–vibrational wavefunctions, energies, term values and vibrationally averaged properties will be presented in a forthcoming publication.

THE MCSCF METHOD

It is essential in a computation of the present scope to obtain correlated electronic wavefunctions and energies, since the correlation energy is expected to be significantly different at different internuclear distances as well as for the different states. It is well known that a configuration interaction (CI) type wavefunction

$$\Psi = \sum_I C_I \Phi_I \qquad (1)$$

is capable of representing correlation effects exactly, provided the CI expansion is carried far enough. It is obvious that the convergence of this CI expansion will depend critically on the appropriateness of the expansion functions Φ_I. In the conventional CI methods as here, the Φ's are configuration state functions (CSF's), i.e., specific linear combinations of Slater determinants (SD's), such that they are eigenfunctions of the symmetry operators of the system. The SD's themselves are constructed from symmetry adapted orbitals. The expansion functions Φ_I and thus the convergence of (1) will depend therefore on the type of orbitals which are used in their construction, as well as on the detailed functional shape of the orbitals. It is here where the difference between conventional CI and MCSCF lies. In conventional CI only the expansion coefficients C_I are optimized variationally, while the MCSCF also optimizes variationally the detailed shape of all the orbitals entering the total wavefunction. Thus one obtains a wavefunction which will give the lowest possible energy with the given number and type of CSF's.

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The procedure for computation is then to

(1) select initial orbitals and configuration types,
(2) solve the CI problem for the state desired,
(3) construct the SCF equations from the first and second order density matrices of the selected state of (2);
(4) solve the SCF equations.

Steps (2)–(4) are iterated to convergence. If we select in (2) always the Kth lowest root, then one will obtain an upper bound to the true energy of the Kth state of a given symmetry. Unfortunately this process does not always work except for the lowest state of a given symmetry. The reason for this is as follows. Since the orbitals are optimized in the field corresponding to state K, this state will be described better and its energy lowered. However if one of the lower states, for instance, state J, has a very different charge distribution than state K, thus requiring quite different orbitals, it follows that the orbitals obtained for state K will be poorer for state J. Thus the energy of state J, originally below state K, will be raised. This may eventually lead to a flipping of the energy order of these two states, preventing convergence of the MCSCF process. Such a state flipping will always happen when the excited state desired has a state of the same symmetry only a little lower in energy, but with a substantially different charge distribution. This is exactly the case for the $A\ ^1\Sigma^+$ state of LiH, the $X\ ^1\Sigma^+$ state being lower by only a few eV. It is possible in this case to optimize the orbitals in the averaged field of both states, obtaining compromise orbitals which will describe both states equally well, however neither optimally. Fortunately it is possible to make up for the deficiency in these compromise orbitals by the addition of a few singly excited configurations. This will become clear in the discussion on configuration selection. The details of the MCSCF method as well as the "averaged field method" are described explicitly elsewhere.[7]

CHOICE OF CONFIGURATIONS

It appears reasonable that in LiH the correlation energy of the 1σ shell, essentially the K shell of lithium, remains nearly constant as the potential curves are traversed and as the outer electrons are excited. Therefore we have chosen to correlate only the outer two valence electrons, which simplifies the problem significantly. In the two electron correlation problem singly excited configurations do not contribute; therefore it is possible to account for the correlation energy of the $B\ ^1\Pi$, $^3\Pi$ and $^3\Sigma^+$ states by using only doubly excited configurations. The particular ones chosen for these three states are presented in Table I, where the bold configuration signifies the dominant one in each state for all internuclear distances calculated.

The configuration choice for the averaged field calculation of $X\ ^1\Sigma^+$ and $A\ ^1\Sigma^+$ is not as straightforward.

TABLE I. Types of valence configurations used.

$X\ ^1\Sigma^+$ and $A\ ^1\Sigma^+$	$1\sigma^2(2\sigma^2+4\sigma^2+5\sigma^2+6\sigma^2+1\pi^2+2\pi^2+1\delta^2+2\delta^2$ $2\sigma 3\sigma+4\sigma 5\sigma+4\sigma 3\sigma+5\sigma 3\sigma+6\sigma 3\sigma+1\pi 2\pi+1\delta 2\delta)$
$B\ ^1\Pi$	$1\sigma^2(\mathbf{2\sigma 1\pi}+3\sigma 2\pi+4\sigma 3\pi+2\pi 1\delta+3\pi 1\delta)$
$^3\Pi$	$1\sigma^2(\mathbf{2\sigma 1\pi}+3\sigma 2\pi+4\sigma 3\pi+2\pi 1\delta+3\pi 1\delta)$
$^3\Sigma^+$	$1\sigma^2(\mathbf{2\sigma 3\sigma}+4\sigma 5\sigma+1\pi 2\pi+1\delta 2\delta)$

Fifteen configurations were used and are given in Table I. The lowest energy configuration of $^1\Sigma^+$ symmetry at intermediate R values is $1\sigma^2 2\sigma^2$, and of necessity ionic (Li^+H^-). The 2σ is a $1s$ H^- type orbital. The configuration of next lowest energy in the intermediate R range is $1\sigma^2 2\sigma 3\sigma$. In this case the 2σ orbital is a much more contracted $1s$ H-like orbital, and the 3σ is a diffuse $2s$–$2p$ hybrid on Li. Because of the double occupation in the ground state, the 2σ orbital determined in the averaged field calculations is essentially that for the ground state. Thus it is too diffuse in nature to describe adequately the 2σ orbital in the $A\ ^1\Sigma^+$ state. The purpose of the thirteen additional configurations was to correlate the ground and first excited state and to make up for the 2σ orbital deficiency in the excited state. Because of the doubly occupied configurations, the 4σ, 5σ, and 6σ orbitals essentially contribute to in–out and left–right correlation of the ground state. By not including a $3\sigma^2$ configuration which would attempt to correlate the ground 2σ also, we tried to make the 3σ orbital an orbital of the excited state only. By considering $2\sigma_A\ ^1\Sigma^+ = 2\sigma_X\ ^1\Sigma^+ + 4\sigma + 5\sigma + 6\sigma$ in order to make up for orbital deficiency, the configurations $3\sigma 4\sigma$, $3\sigma 5\sigma$, and $3\sigma 6\sigma$ arise. The doubly occupied π and δ configurations primarily introduce angular correlation into the ground state, while the $1\pi 2\pi$ and $1\delta 2\delta$ configurations are more effective in correlating the $A\ ^1\Sigma^+$ state. A discussion of the orbitals and important configurations in the wavefunctions of these two $^1\Sigma^+$ states at several internuclear distances is postponed until the next section.

It was found both convenient and time saving to move across the potential curves in one direction using the converged orbital coefficients for each state at the preceding internuclear distance to begin the calculation at each point. No convergence difficulties in solving the SCF equations were encountered in the calculation of the $B\ ^1\Pi$, $^3\Pi$ and $^3\Sigma^+$ states. These states retained the same dominant configuration all across the potential curves, the orbitals of the additional configurations being solely used to correlate the valence electrons. In the averaged field calculation of the $A\ ^1\Sigma^+$ and $X\ ^1\Sigma^+$ states, convergence difficulties, characterized mainly by orbital flipping among the valence sigma orbitals, could be avoided when the R step size through the curve-crossing regions was made small enough. It is

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TABLE II. Basis set of Slater-type orbitals: $23\sigma \times 8\pi \times 4\delta$.

		N	L	M	K	Zeta
1	1	1	0	0	1	4.6351
2	2	1	0	0	1	2.4730
3	3	2	0	0	1	1.0330
4	4	2	0	0	1	0.8237
5	5	2	0	0	1	0.5100
6	6	3	0	0	1	2.6811
7	7	2	1	0	1	3.9004
8	8	2	1	0	1	2.1109
9	9	2	1	0	1	1.0758
10	10	2	1	0	1	0.7359
11	11	2	1	0	1	0.5100
12	12	2	1	0	1	0.3500
13	13	3	2	0	1	1.4974
14	14	3	2	0	1	0.9866
15	15	4	3	0	1	1.7748
16	16	1	0	0	2	1.9883
17	17	1	0	0	2	1.0000
18	18	1	0	0	2	0.7000
19	19	1	0	0	2	0.4000
20	20	2	0	0	2	2.2425
21	21	2	1	0	2	1.0638
22	22	2	1	0	2	0.5480
23	23	3	2	0	2	1.5774
24	1	2	1	1	1	1.5600
25	2	2	1	1	1	0.7800
26	3	2	1	1	1	0.5472
27	4	2	1	1	1	0.3500
28	5	3	2	1	1	1.0000
29	6	2	1	1	2	2.0000
30	7	2	1	1	2	1.0000
31	8	3	2	1	2	2.1000
32	1	3	2	2	1	1.0000
33	2	3	2	2	1	0.5000
34	3	3	2	2	2	2.0000
35	4	3	2	2	2	1.0000

felt that the dual role of these orbitals—that of correlating the ground state 2σ and making up for deficiencies in the excited state 2σ—imposed a strain. The valence sigma orbitals were attempting to optimize simultaneously in several different regions of space.

CHOICE OF BASIS SET

The importance of choosing a good basis set cannot be underestimated in a multiconfiguration technique, for, in the limit that all configurations are included, the basis set determines the quality of the calculation. Since the same basis set of Slater-type functions (STF's) was used for four of the five states of LiH, the following factors were considered important in selecting it:

(1) how well it reproduced the ground state Hartree–Fock

(2) how adequate it was for description of correlating orbitals

(3) in the averaged field case, how it could compensate for orbital deficiencies

(4) how well it could describe the excited states.

Utilizing the same basis set for all states is economical, in that integrals have to be calculated only once at each internuclear distance. Also, it ensures that the representation of the core orbital does not change from one state to another. The calculation of the transition moment, in addition, is considerably simplified. Since one of our primary interests was in accurately calculating the ground and first excited $^1\Sigma^+$ states, we knew that we needed a large and flexible basis set. It seemed reasonable, therefore, that with few extra functions added the basis would be sufficient for the Π states also.

The basis set used in our final calculation is listed in Table II. A subset of 21σ, 7π and 4δ STF's (all the listed functions except for those numbered 12, 19, 27) was optimized at the equilibrium internuclear distance for the $X\,^1\Sigma^+$ state with the 2σ shell correlated. Preliminary calculations were carried out with this reduced basis and it was found necessary to add the three additional diffuse functions (12, 19, and 27) in order to describe better the $A\,^1\Sigma^+$ state and the Π states. The $^3\Sigma^+$ state, however, was not recalculated. The total basis set yielded for the $X\,^1\Sigma^+$ state the Hartree–Fock energy of -7.987317 hartree at 3.0 bohr, essentially the Hartree–Fock limit.

CALCULATED POTENTIAL CURVES

The potential curves obtained are given in Tables III and IV and displayed in Fig. 1. Within the framework of the Hartree–Fock model the $X\,^1\Sigma^+$ state with configuration $1\sigma^2 2\sigma^2$ will dissociate to ions, $Li^+(^1S)+$

TABLE III. Potential curves for four states of LiH.

R (bohrs)	$X\,^1\Sigma^+$, E(hartrees)	$A\,^1\Sigma^+$, E(hartrees)	$B\,^1\Pi$, E(hartrees)	$^3\Pi$, E(hartrees)
2.00	−7.948461	−7.804494	−7.7667212	−7.7857226
2.25	−7.986659	−7.841651	−7.8048017	−7.8225261
2.50	−8.007771	−7.865183	−7.8288091	−7.8452320
2.75	−8.017994	−7.880054	−7.8438361	−7.8588998
3.00	−8.021254	−7.889358	−7.8531226	−7.8667765
3.25	−8.020163	−7.895156	−7.8587516	−7.8709706
3.50	−8.016339	−7.898690	−7.8620695	−7.8728632
4.00	−8.004649	−7.902121	−7.8649605	−7.8730600
4.50	−7.990950	−7.903239	−7.8656627	−7.8714465
5.00	−7.977741	−7.903563	−7.8656819	−7.8696445
5.50	−7.965892	−7.903427	−7.8655471	−7.8681780
6.00	−7.955901	−7.902660	−7.8654140	−7.8671223
6.50	−7.948031	−7.900999	−7.8653134	−7.8664058
7.50	−7.938476	−7.894651	−7.8651918	−7.8656261
8.50	−7.934672	−7.885984	−7.8651279	−7.8652968
10.00	−7.933082	−7.874234	−7.8650787	−7.8651187
12.00	−7.932752	−7.866665	−7.8650508	−7.8650565

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$H^-(^1S)$. The ionic dissociation limit lies about 4.64 eV above the neutral atoms. [The ionization potential of $Li(^2S)$ is 5.39 eV[8] and the electron affinity of $H(^2S)$ is taken as 0.75 eV[9].] The neutral, ground-state atoms $Li(^2S)$ and $H(^2S)$ can give rise, however, to molecular states $^1\Sigma^+$ and $^3\Sigma^+$. By the adiabatic noncrossing rule, therefore, the $X\,^1\Sigma^+$ state must dissociate to $Li(^2S)+H(^2S)$. Consequently the wavefunction must change character from ionic to neutral as the potential curve is traversed from small to large R, and this is well described by an MCSCF type wavefunction.

The $A\,^1\Sigma^+$ state, with molecular configuration $1\sigma^2 2\sigma 3\sigma$, is predominantly Li^-H^+ for small R. In the region 6.0–10.0 bohr, however, this state interacts strongly with the $1/R$ ionic curve which is dominant in the ground state for small R. As the ground state becomes more neutral in order to dissociate correctly,

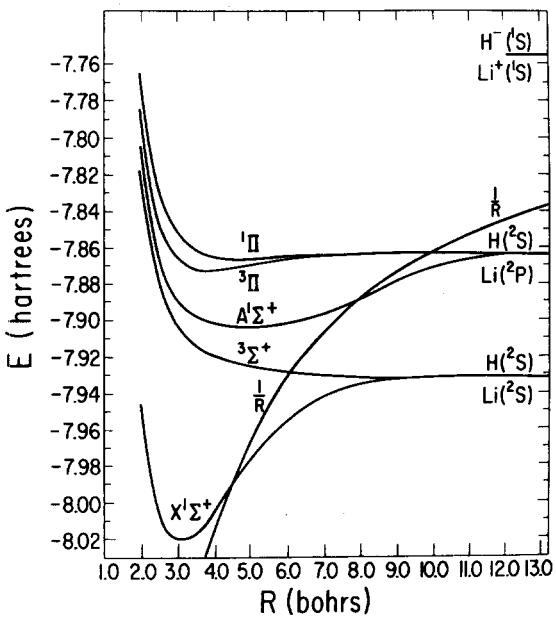


FIG. 1. Potential curves of five states of LiH. Included also is a $1/R$ curve, referred to the ionic limit $Li^+ + H^-$, $E = -7.764$ hartree, at $R = \infty$.

the $A\,^1\Sigma^+$ becomes ionic (Li^+H^-) with dominant configuration $1\sigma^2 2\sigma^2$. This state would dissociate to ions if it were not for the fact that the excited atoms $Li(^2P)+H(^2S)$ can also give rise to a $^1\Sigma^+$ state. This dissociation limit lies 1.85 eV above the ground state neutrals—almost three electron volts lower than the ionic limit. Therefore, the $A\,^1\Sigma^+$ state must for large R become neutral and dissociate to $Li(^2P)+H(^2S)$. Two avoided crossings which alter the character of the wave function occur in this $A\,^1\Sigma^+$ state.

In Table V are listed the four most important configurations in the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ calculation and the appropriate CI mixing coefficients for each state at three different internuclear distances. These distances, 3.0, 7.5, and 12.0 bohr were chosen to show how the

TABLE IV. Potential curve points for $^3\Sigma^+$.[a]

R (bohrs)	Energy (hartrees)	R (bohrs)	Energy (hartrees)
2.0	−7.81863	4.0	−7.91984
2.5	−7.87889	5.0	−7.92584
2.9	−7.90027	6.0	−7.92944
3.0	−7.90366	7.5	−7.93189
3.1	−7.90655	9.0	−7.93256
3.5	−7.91442	12.0	−7.93271

[a] Calculated using a $21\sigma \times 7\pi \times 4\delta$ set of STO's given in Table III eliminating basis functions 12, 19, and 27.

character of the wavefunctions for these states changes with R. The orbital expansion coefficients for the 2σ, 3σ, and 4σ orbitals at the three internuclear distances are presented in Tables VI, VII, and VIII, respectively. At 3.0 bohr the dominant configuration for the ground state is $1\sigma^2 2\sigma^2$ with the diffuse 2σ having most weight on basis functions 17, 18, and 19 (Table VI). There is some small Li $2s$–$2p$ contribution—enough to indicate that this is a molecular orbital. The 3σ is diffuse $2s_{Li}$ with some $2p_{Li}$ and diffuse H^- character—also a *molecular* orbital. The 4σ character is difficult to pinpoint, but it has large contributions on H basis functions to correlate the 2σ and contributions arising also from Li $2s$ and $2p$ functions.

In going from 3.0 to 7.5 bohr the orbitals of course do change character slightly, but not drastically. Individual changes are difficult to identify with such a large basis merely by looking at the orbital expansion coefficients. The change in the configuration mixtures for the states is to be noted at 7.5 bohr. Now the $1\sigma^2 2\sigma 3\sigma$ configuration is dominant in the $X\,^1\Sigma^+$ state and the $A\,^1\Sigma^+$ state is predominantly $1\sigma^2 2\sigma^2$. The ground state has become neutral and the first excited state ionic.

TABLE V. Dominant configurations in the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states.

	Configuration mixing coefficients			
	$1\sigma^2 2\sigma^2$	$1\sigma^2 2\sigma 3\sigma$	$1\sigma^2 4\sigma^2$	$1\sigma^2 3\sigma 4\sigma$
$R = 3.0$ bohr				
$X\,^1\Sigma^+$	0.96	−0.23	−0.10	0.07
$A\,^1\Sigma^+$	0.24	0.94	−0.03	−0.24
$R = 7.5$ bohr				
$X\,^1\Sigma^+$	−0.40	0.78	0.13	−0.39
$A\,^1\Sigma^+$	0.84	0.44	−0.22	−0.21
$R = 12.0$ bohr				
$X\,^1\Sigma^+$	−0.07	0.77	0.04	−0.62
$A\,^1\Sigma^+$	0.84	0.06	−0.54	−0.05

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TABLE VI. Comparison of 2σ orbital for X and $A\ ^1\Sigma^+$ states at three internuclear distances.

	σ Basis functions					Orbital coefficients		
	N	L	M	K	Zeta	3.0 bohr	7.5 bohr	12.0 bohr
1	1	0	0	1	4.6351	−0.00305	−0.00045	−0.00058
2	1	0	0	1	2.4730	−0.12211	−0.01359	−0.00412
3	2	0	0	1	1.0330	−0.01111	−0.14824	−0.01674
4	2	0	0	1	0.8237	0.20176	0.24952	−0.01999
5	2	0	0	1	0.5100	−0.17579	−0.12169	0.04033
6	3	0	0	1	2.6811	−0.03982	0.00632	−0.00305
7	2	1	0	1	3.9004	0.00560	0.00166	0.00103
8	2	1	0	1	2.1109	−0.00410	0.00763	0.00783
9	2	1	0	1	1.0758	0.09521	−0.03667	0.02540
10	2	1	0	1	0.7359	0.11922	0.12243	−0.03274
11	2	1	0	1	0.5100	0.03258	0.24619	0.61914
12	2	1	0	1	0.3500	−0.01408	−0.05585	−0.02548
13	3	2	0	1	1.4974	0.00919	−0.00906	−0.00143
14	3	2	0	1	0.9866	0.01120	0.02561	0.00347
15	4	3	0	1	1.7748	0.00279	0.00094	0.00013
16	1	0	0	2	1.9883	0.07231	0.07333	0.01961
17	1	0	0	2	1.0000	0.26934	0.19550	0.57531
18	1	0	0	2	0.7000	0.40755	0.39494	0.03894
19	1	0	0	2	0.4000	0.13425	0.24966	0.16892
20	2	0	0	2	2.2425	0.01403	0.02760	0.00469
21	2	1	0	2	1.0638	−0.02485	0.01042	0.00765
22	2	1	0	2	0.5480	−0.02622	−0.05529	−0.01872
23	3	2	0	2	1.5774	0.00344	0.00153	−0.00012

At 12.0 b the 2σ orbital is well represented as a linear combination of a diffuse Li $2p$ function and an H $1s$, i.e., $(2p_{\mathrm{Li}}+1s_{\mathrm{H}})$. The 4σ orbital is just the orthogonal component to this, $(2p_{\mathrm{Li}}-1s_{\mathrm{H}})$. Very atomiclike at this internuclear distance, the 3σ orbital is $2s_{\mathrm{Li}}$ with essentially no $2p_{\mathrm{Li}}$ character. Because the $A\ ^1\Sigma^+$ dissociates to $\mathrm{Li}(^2P)+\mathrm{H}(^2S)$ and has no Li $2s$ character at large R, the weights for configurations $1\sigma^2 2\sigma 3\sigma$ and $1\sigma^2 3\sigma 4\sigma$ are very small in this state. One can understand the configuration mixture for both states from a valence-bond standpoint. For the $A\ ^1\Sigma^+$ state the $2\sigma^2$ configuration is essentially $(2p_{\mathrm{Li}}+1s_{\mathrm{H}})^2$ while the $4\sigma^2$ configuration $(2p_{\mathrm{Li}}-1s_{\mathrm{H}})^2$ enters with the opposite sign. Thus valence bond components $2p_{\mathrm{Li}}1s_{\mathrm{H}}$ are enhanced, while the ionic components $2p_{\mathrm{Li}}^2$ and $1s_{\mathrm{H}}^2$ are subtracted. The $2p_{\mathrm{Li}}1s_{\mathrm{H}}$ description is necessary to be consistent with the dissociation limit.

The same sort of argument explains the equal magnitude but opposite sign mixture of $2\sigma 3\sigma$ and $3\sigma 4\sigma$ in the $X\ ^1\Sigma^+$ state. We have, in terms of atomic functions $(2p_{\mathrm{Li}}+1s_{\mathrm{H}})(2s_{\mathrm{Li}})-(2p_{\mathrm{Li}}-1s_{\mathrm{H}})(2s_{\mathrm{Li}})$. The $2p_{\mathrm{Li}}2s_{\mathrm{Li}}$ contributions cancel, while the $2s_{\mathrm{Li}}1s_{\mathrm{H}}$ components add, giving us what we expect for dissociation to $\mathrm{Li}(^2S)+\mathrm{H}(^2S)$.

The remaining states arising from the $\mathrm{Li}(^2P)+\mathrm{H}(^2S)$ separated atom limit are $B\ ^1\Pi$, $^3\Pi$ and $^3\Sigma^+$. Of these, we performed calculations on the Π states, only, which have the dominant configuration $1\sigma^2 2\sigma 1\pi$. The $^3\Sigma^+$ state we did compute is the lowest of that particular symmetry (with configuration $1\sigma^2 2\sigma 3\sigma$) and is a repulsive state dissociating to ground state neutrals.

In order to calculate dissociation energies from the potential curves for each of the molecular states we need the energies of the separated atoms at the dissociation limits. Because we are not correlating the 1σ lithium core at all we want the energy of the separated atoms to reflect this. Therefore we choose to take for the reference energies at the dissociation limit $\mathrm{Li}(^2S)+\mathrm{H}(^2S)$ the calculated Hartree–Fock energy of $\mathrm{Li}(^2S)$, $E=-7.432726$ hartree[10] and the exact energy for $\mathrm{H}(^2S)$ of $E=-0.5$ hartree. The sum of these two energies, $E=-7.932726$ hartree, gives the separated atom ground state energy with an uncorrelated $1s$ shell on lithium. The $X\ ^1\Sigma^+$ and $^3\Sigma^+$ state dissociate to these separated atoms. It can be seen from Tables III and IV that at 12.0 bohr both states are virtually at this energy limit.

The second dissociation limit of interest is $\mathrm{Li}(^2P)+\mathrm{H}(^2S)$, involving a $2s\rightarrow 2p$ excitation of the lithium valence electron from the ground state atom. Using the $\mathrm{Li}(^2P)$ Hartree–Fock energy, $E=-7.365068$ hartree,[11] the dissociation limit energy for $\mathrm{Li}(^2P)+\mathrm{H}(^2S)$ is -7.865068 hartree. An estimate for this limit could also be made using the $\mathrm{Li}(^2S)$ Hartree–Fock and the $2s\rightarrow 2p$ excitation energy.[8] An energy about 73 cm^{-1} higher than that quoted above is obtained because this does not account for a difference in correlation in the $1s^2 2s$ versus $1s^2 2p$ configurations. It

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TABLE VII. Comparison of 3σ orbital for X and $A\ ^1\Sigma^+$ states at three internuclear distances.

	σ Basis functions					Orbital coefficients		
	N	L	M	K	Zeta	3.0	7.5	12.0
1	1	0	0	1	4.6351	0.00966	0.01023	0.01039
2	1	0	0	1	2.4730	0.10867	0.15640	0.15903
3	2	0	0	1	1.0330	0.24515	0.41298	0.47360
4	2	0	0	1	0.8237	−0.70832	−1.09659	−1.19227
5	2	0	0	1	0.5100	−0.71703	−0.36182	−0.29639
6	3	0	0	1	2.6811	0.01010	0.02160	0.01323
7	2	1	0	1	3.9004	0.00135	0.00024	0.00015
8	2	1	0	1	2.1109	0.01054	0.00477	0.00077
9	2	1	0	1	1.0758	−0.01736	−0.00845	0.00595
10	2	1	0	1	0.7359	0.19866	0.02614	−0.01513
11	2	1	0	1	0.5100	0.29576	0.20617	0.09802
12	2	1	0	1	0.3500	−0.06852	−0.05162	−0.01816
13	3	2	0	1	1.4974	0.00091	−0.00326	−0.00020
14	3	2	0	1	0.9866	0.00420	0.00895	0.00050
15	4	3	0	1	1.7748	0.00012	0.00025	0.00002
16	1	0	0	2	1.9883	0.01686	0.00356	−0.00255
17	1	0	0	2	1.0000	−0.05517	0.00899	0.02582
18	1	0	0	2	0.7000	−0.02557	−0.06606	−0.04177
19	1	0	0	2	0.4000	0.44309	0.16701	0.03730
20	2	0	0	2	2.2425	0.00412	0.00049	−0.00183
21	2	1	0	2	1.0638	0.00393	−0.00130	−0.00176
22	2	1	0	2	0.5480	−0.03627	−0.02082	−0.00494
23	3	2	0	2	1.5774	0.00062	0.00062	0.00022

may be seen from Table III that the $^3\Pi$ and $B\ ^1\Pi$ states have converged to this limit at 12.0 bohr and that the $A\ ^1\Sigma^+$ state is still slightly below this limit.

The energy minima of the potential curves for the four bound states were obtained by polynomial interpolation between the calculated points and are listed in Table IX. Also listed in this table are the internuclear distance at the energy minimum ($R_{\min}$), the dissociation energies calculated from $E_{\min}$ and the dissociation limits discussed here, and the experimental values. The calculated D_e's are all less than the experimental quantities which is to be expected in a variational type calculation where the exact energy eigenvalue has not yet been reached.

The experimental D_e's are thought to be relatively accurate—to within ± 0.0002 eV. The bands observed by Velasco[6] for the $B\ ^1\Pi$–$X\ ^1\Sigma^+$ system showed clear breaking off of the rotational structure. This was attributed to rotational predissociation of the upper state. Extrapolation of the limiting curve of dissociation yields a dissociation limit for the $B\ ^1\Pi$ state which is felt to be accurate to within several wavenumbers. Using the $2s \rightarrow 2p$ excitation energy, the dissociation limit for the ground state was obtained by Velasco also.

The largest difference between our calculated and the experimental D_e is in the $X\ ^1\Sigma^+$ state. The calculated D_e is about 845 cm^{-1} smaller than experiment would indicate. Thus the minimum of our potential curve for this state lies 845 cm^{-1} too high. This discrepancy in the ground state is not surprising for several reasons. One factor is the use of the averaged field. At 3.0 bohr, using the same set of configurations as in the averaged field calculation, the energy for the $X\ ^1\Sigma^+$ state was obtained alone. This energy, $E = -8.021974$ hartree, is 143 cm^{-1} below the averaged field $X\ ^1\Sigma^+$ minimum. Another source of error is neglecting to account for the change in the intershell correlation. As the potential curve is traversed from small to large R, the $2\sigma^2$ pair breaks apart, leaving at the separated atom limit only two 1σ–2σ (or $1s$–$2s$) pairs instead of four. At the potential minimum, therefore, we have twice as much correlation error due to 1σ–2σ pairs as at the separated limit. This is estimated, from calculations on the K–L shell correlation energy of the united atom Be,[12] to be not more than 500 cm^{-1}. The largest error, however, is probably due to incomplete correlation of the 2σ shell. We estimate, from calculations of comparable accuracy using the MCSCF method for two- and four-electron atomic systems,[12] that 90%–95% of the correlation energy of the 2σ pair was obtained. Therefore somewhere between 400 and 800 cm^{-1} of correlation energy is still unaccounted for. Our error of 845 cm^{-1} is thus easily attributed to a combination of the three factors just discussed.

The other two states for which experimental D_e's are known, the $B\ ^1\Pi$ and $A\ ^1\Sigma^+$, are shown to be too shallow by 143 and 232 cm^{-1}, respectively. Although we have

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TABLE VIII. Comparison of 4σ orbital for X and $A\ ^1\Sigma^+$ states at three internuclear distances.

	σ Basis functions					Orbital coefficients		
	N	L	M	K	Zeta	3.0	7.5	12.0
1	1	0	0	1	4.6351	0.00654	−0.00154	−0.00061
2	1	0	0	1	2.4730	−0.14538	−0.02034	−0.00513
3	2	0	0	1	1.0330	0.14033	−0.29442	0.02668
4	2	0	0	1	0.8237	0.14749	0.48290	−0.03019
5	2	0	0	1	0.5100	−0.45533	−0.24050	0.05143
6	3	0	0	1	2.6811	−0.07510	0.01469	−0.00504
7	2	1	0	1	3.9004	−0.00510	−0.00577	0.00365
8	2	1	0	1	2.1109	0.01181	0.03837	0.00348
9	2	1	0	1	1.0758	0.11971	−0.13880	0.04712
10	2	1	0	1	0.7359	0.39131	0.34243	−0.06016
11	2	1	0	1	0.5100	−0.04435	0.38885	0.78293
12	2	1	0	1	0.3500	0.02392	−0.07891	−0.03383
13	3	2	0	1	1.4974	0.01238	−0.01748	−0.00105
14	3	2	0	1	0.9866	−0.00931	0.04821	0.00263
15	4	3	0	1	1.7748	−0.00335	0.00154	0.00009
16	1	0	0	2	1.9883	−0.01101	0.09821	0.02109
17	1	0	0	2	1.0000	−2.65779	−1.80925	−0.88332
18	1	0	0	2	0.7000	2.35030	0.89112	0.05627
19	1	0	0	2	0.4000	0.21902	0.44637	0.20362
20	2	0	0	2	2.2425	−0.16867	0.00603	0.00403
21	2	1	0	2	1.0638	−0.24561	0.00745	0.01779
22	2	1	0	2	0.5480	−0.07226	−0.11000	−0.02214
23	3	2	0	2	1.5774	0.01260	0.00286	−0.00078

made the least absolute error in calculating the $B\ ^1\Pi$ state, the 143 cm^{-1} represents about half the binding energy for this state and thus will greatly affect subsequent calculations of spectroscopic quantities. The correlation energy in these two states is very small compared to that in the $X\ ^1\Sigma^+$ state—less than 10%. The remaining correlation energy between the valence electrons unaccounted for is felt to be less than 50 cm^{-1}. The amount by which our dissociation energies differ from experiment probably reflects the neglect of the change in correlation of two 1σ–2σ pairs as the potential curve is traversed.

The calculated T_e values evidence, also, the fact that the error in the ground state is much larger than the error in either excited state:

$$T_e(A\ ^1\Sigma^+ - X\ ^1\Sigma^+) = 25\,842\ cm^{-1} \quad \text{(exptl 26 510)}$$

$$T_e(B\ ^1\Pi - X\ ^1\Sigma^+) = 34\,153\ cm^{-1} \quad \text{(exptl 34 912)}$$

$$T_e(A\ ^1\Sigma^+ - B\ ^1\Pi) = 8\,310\ cm^{-1} \quad \text{(exptl 8 402)}$$

The R_{min} for the $X\ ^1\Sigma^+$ state agrees reasonably well with experiment; however, the agreement for the $A\ ^1\Sigma^+$ and $B\ ^1\Pi$ states appears exceptionally poor. The reason for this may be seen in the fact that both of these po-

TABLE IX. Energy minima, dissociation energies, and R_e's for four states of LiH.

State	E_{min}(hartrees)[a]	D_e(calc, eV)	D_e(exptl, eV)[b]	R_{min}(bohrs)[a]	R_e(exptl, bohrs)
$X\ ^1\Sigma^+$	−8.021321	2.411[c]	2.5154	3.05	3.015[d]
$A\ ^1\Sigma^+$	−7.903574	1.048[e]	1.0765	5.12	4.9064[d]
$B\ ^1\Pi$	−7.865709	0.017[e]	0.035	4.75	4.494[b]
$^3\Pi$	−7.873358	0.226[e]		3.76	

[a] Minimum of electronic potential curve obtained by fifth order polynomial interpolation.
[b] See Ref. 6.
[c] Dissociation limit energy of $Li(^2S)+H(^2S)$, $E=-7.932726$ hartree.
[d] See Ref. 9, p. 546.
[e] Dissociation limit energy of $Li(^2P)+H(^2S)$, $E=-7.865068$ hartree.

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tential curves are very flat around their minima. It will be seen, however, in a following paper presenting a spectroscopic analysis of the potential curves, that the discrepancy is also due to the fact that R_e values obtained by an extrapolation of the B_v's do not correspond to the minima of these potential curves.

DISCUSSION AND COMPARISON WITH OTHER THEORETICAL RESULTS

Lithium hydride has been the testing ground for almost every *ab initio* method proposed for extending calculations beyond the Hartree–Fock level in molecules; and although a plethora of theoretical calculations on LiH exist, there are very few for which a comparison with ours is instructive. This is due to the fact that most calculations are for the ground state at its potential minimum ($R_e=3.015$ bohr), only, and most attempt correlation of both core and valence electrons. A brief review of several of these calculations follows. For comparison, Cade and Huo's calculation ($E=-7.987313$ hartree) for the $X\,^1\Sigma^+$ state is probably close to the true Hartree–Fock energy for the system. The experimental energy at the $X\,^1\Sigma^+$ potential minimum is -8.0703.[13] Although not an upper bound, the lowest energy obtained to date for the ground state is that of Boys and Handy[14] ($E=-8.063$ hartree) using a transcorrelated wavefunction containing terms explicitly depending on r_{ij}. The best variational calculation was done by Bender and Davidson,[15] using a natural orbital approach and a set of elliptic basis functions. They obtained an energy minimum of -8.0606 hartree. Many other results using various techniques could be mentioned[16–27]; however, we will compare our work only with those calculations which correlate the valence shell.

Our MCSCF method, correlating the valence shell alone is akin to Wahl's OVC method.[28] Mukherjee and McWeeny,[29] using a similar approach with a frozen K shell, chose nine configurations to correlate the 2σ valence shell in the ground state, obtaining an energy of -8.01488 hartree at R_e. As our basis set was almost three times as large as theirs, we expect to do significantly better for the $X\,^1\Sigma^+$ state; hence our value of $E=-8.02131$ hartree is not surprising.

To date, the most comprehensive study of all the potential curves which we investigated is that of Bender and Davidson,[30] who used a frozen 1σ core and limited configuration interaction to obtain nineteen states at nine different internuclear distances. Their work represents the only other calculation of the $B\,^1\Pi$ and $^3\Pi$ potential curves besides our present results, and Bender and Davidson do not obtain a bound $B\,^1\Pi$. At $R_e=3.0156$ they obtain an energy of -8.0036 hartree for the $X\,^1\Sigma^+$ state. Our value lies almost 3900 cm^{-1} lower. The main difference here is thought to be our optimization of the correlating orbitals. In preliminary calculations, using the Bender–Davidson basis set of 13σ, 4π, and 1δ STF's, we obtained for the $X\,^1\Sigma^+$ state with four MCSCF configurations $1\sigma^2(2\sigma^2+3\sigma^2+4\sigma^2+1\pi^2)$ an energy of $E=-8.0173$ hartree; 3000 cm^{-1} lower than their fifty configuration result. To investigate whether our energy improvement might be due to changes in the K shell upon correlation of the valence shell, we froze our 1σ orbital to the Hartree–Fock value and carried out the same calculation. The energy change was only 6 cm^{-1}, showing that the K shell is affected very little by correlation of the valence shell. Thus the 3000 cm^{-1} difference between the 50 configuration CI result and our four configuration MCSCF result is due to our optimization of the valence and correlating orbitals only. Much of the remaining difference between our present calculation and that of Bender and Davidson is due to our more extensive basis set.

In the $A\,^1\Sigma^+$ curve, Bender and Davidson calculate the minimum to be $E=-7.8979$ hartree compared to our -7.9036 hartree—a difference of $\sim$1250 cm^{-1}. In their calculation they obtain a repulsive $B\,^1\Pi$ state. The lowest energy which they obtain is at 6.0 bohr, $E=-7.8606$ hartree, compared to our minimum for this state which is at least 1000 cm^{-1} below this value, and at $R\approx 4.7$ bohr. The minimum of their $^3\Pi$ curve, $E=-7.8666$ hartree, is again higher than ours by about 1500 cm^{-1}. The fact that the $^3\Pi$ state is more poorly described than the $B\,^1\Pi$ in Bender and Davidson's calculation is probably due to an inadequate π basis set on H. The $^3\Pi$ state has a dipole moment Li^+H^- which indicates a small shift of charge to H occurring in the 1π orbital.

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LiH Properties, Rotation–Vibrational Analysis, and Transition Moments for $X\ ^1\Sigma^+$, $A\ ^1\Sigma^+$, $B\ ^1\Pi$, $^3\Sigma^+$, and $^3\Pi$*

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Using accurate *ab initio* calculated potential curves and electronic wavefunctions for the states $X\ ^1\Sigma^+$, $A\ ^1\Sigma^+$, $B\ ^1\Pi$, $^3\Sigma^+$, and $^3\Pi$ for LiH, various properties were calculated. These include dipole and quadrupole moment, field gradient at the nuclei, etc. Rotation–vibrational wavefunctions were obtained and a rotation–vibrational analysis was carried out. Some of the properties obtained were averaged over the appropriate rotation–vibrational wavefunctions. In addition electronic transition moments were computed and from this uv and ir line strengths were obtained. In general the agreement with experimental values, where available, is satisfactory.

In the preceding paper, hereafter referred to as Paper I,[1] we discussed the calculation of the potential curves for five states of LiH: $X\ ^1\Sigma^+$, $A\ ^1\Sigma^+$, $B\ ^1\Pi$, $^3\Pi$, and $^3\Sigma^+$. In computing electronic energies and wavefunctions, essentially one-half of the Born–Oppenheimer problem has been solved. Now, not only can we solve the second half of the Born–Oppenheimer problem for the motion of the nuclei in each bound state, but we can also obtain information about stationary state properties and transition probabilities. The following sections describe the calculation of various molecular properties using the multiconfiguration self-consistent field (MCSCF) wavefunctions and potential curves of Paper I and present the results. These properties are of three types and will be discussed separately: (1) spectroscopic quantities obtained by solution of the one-dimensional radial Schrödinger equation for the nuclear motion; (2) the expectation values of certain one-electron operators evaluated over the electronic wavefunctions of each state; (3) transition moments between different electronic states.

I. VIBRATION–ROTATION ANALYSIS

In the Born–Oppenheimer approximation, it is assumed that the total wavefunction of a diatomic molecule can be expressed as a product of nuclear and electronic wavefunctions, which are solutions of separate equations. The electronic energies and wavefunctions are usually obtained as in Paper I in a field of fixed nuclei and thus depend parametrically on the internuclear distance R. With the electronic energy $U(R)$ playing the role of the potential energy, the Schrödinger equation for the nuclear motion of a diatomic molecule, regarded as a symmetric top, can be separated into angular and radial parts. The solution has the following form:

$$\Psi = R^{-1} P_{v,J}(R)\, Y_{J\Lambda M}(\theta, \varphi, \chi), \tag{1}$$

where the $Y_{J\Lambda M}$'s are the eigenfunctions of the symmetric top.

In the above expression, θ is the angle of the figure axis of the top with a fixed z axis; φ is the azimuthal angle about the z axis; χ is the azimuthal angle measuring rotation about the figure axis (z axis). The quantum number M is the projection of the total angular momentum along an arbitrary axis in space. The quantum numbers J and Λ differ in meaning according to the particular coupling case and will be discussed below. $P_{v,J}(R)$, the vibrational wavefunction, is the solution of the one-dimensional radial Schrödinger equation (in atomic units):

$$(d^2/dR^2) P_{v,J}(R) - 2\mu\{U(R) + [J(J+1) - \Lambda^2]/2\mu R^2 - E_{v,J}\} P_{v,J}(R) = 0, \tag{2}$$

where μ is the reduced mass of the nuclei; $U(R)$ is the electronic energy for *fixed* internuclear distance R (including the Coulomb repulsion of the nuclei), and v is the vibrational quantum number. $U(R)$ together with the centrifugal potential term gives an effective

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